Homework Section 14.4

1. Find an equation of the tangent plane to the surface at the given point:

   a) $z = 4x^2 - y^2 + 2y$, $(-1, 2, 4)$ b) $z = y \ln x$, $(1, 4, 0)$

   c) $z = e^{x^2 - y^2}$, $(1, -1, 1)$

2. Explain why the function is differentiable at the given point. Then find the linearization $L(x, y)$ of the function at that point.

   a) $f(x, y) = x\sqrt{y}$, $(1, 4)$ b) $f(x, y) = e^x \cos xy$, $(0,0)$

3. Find the linear approximation of the function $f(x, y) = \sqrt{20 - x^2 - 7y^2}$ at (2,1) and use it to approximate $f(1.95, 1.08)$.

4. Find the linear approximation of the function $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$ at (3,2,6) and use it to approximate the number $\sqrt{3.02^2 + 1.97^2 + 5.99^2}$. (**Note**: the equation of the tangent plane generalizes to any number of variables.)

5. Find the total differential of the function. For part (b), note that the total differential generalizes to functions of more than two variables.

   a) $z = x^3 \ln y^2$ b) $w = xye^{xz}$

6. Suppose wave heights, $h = f(v, t)$, as a function of wind speed v and duration t are given by the following chart (recorded in feet).

hours

	t \ v	**5**	**10**	**15**
Wind speed (knots)	**10**	2	3	3
	20	8	8	9
	30	12	14	15

Use the table to find a linear approximation to the wave height function when v is near 20 knots and t is near 10 hours.